Problem 2.

(a)

the truth table shows that the $(P \wedge \neg Q) \vee (P \wedge Q)$ is equivalent to P.

P	Q	$\neg Q$	$P \wedge \neg Q$	$P \wedge Q$	$(P \wedge \neg Q) \vee (P \wedge Q)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

(b)

$(A - B)$ is equivalent to $A \wedge \neg B$

the truth table about $(A - B) \vee (A \wedge B)$ is

A	B	$\neg B$	$A \wedge \neg B$	$A \wedge B$	$(A - B) \vee (A \wedge B)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

we can see the A is equivalent to $(A - B) \vee (A \wedge B)$

so $x \in A \iff x \in ((A - B) \vee (A \wedge B))$

Problem 3.

(a) sixish$(n) := \exists m \exists k.((6 \cdot 10^k + m = n) \wedge m < 10^k)$

(b) $\exists m \exists n \forall k.Q(m) \wedge Q(n) \wedge k \neq m \wedge k \neq n \wedge \neg Q(k)$

(c) $\forall k \exists n \exists m.(isEven(k) \wedge k > 2) \implies (m + n = k) \wedge isPrime(n) \wedge isPrime(m)$

(d) $\forall n.n > 100 \implies$ sixish(n)

(e) it is not equivalent because when n = 1, R(n) should be True but we can't find any b $\in \mathbb{N}$ to make 1 = 2b + 5

Problem 4.

(a)

We use induction.

Proof. Let $P(n)$ be the statement that if $Q(x_1 \cdot x_2 \cdots x_n)$ is true, then at least one of $Q(x_1)$, $Q(x_2)$, $\ldots$, $Q(x_n)$ must be true.

Base case: For $n = 1$, the statement holds trivially.For $n = 2$, if $Q(x_1 \cdot X_2)$ is true, then at least one of $Q(x_1)$ or $Q(x_2)$ must be true which holds by the given premise.

Inductive hypothesis: Assume that $P(n)$ is true for some $n \geq 1$. That is, if $Q(x_1 \cdot x_2 \cdots x_n)$ is true, then at least one of $Q(x_1), \ldots, Q(x_n)$ must be true.

Inductive step: We assume that for $n \geq 1$, P(n) is true. We have to prove $P(n+1)$ is true, which means we have to prove if $Q(x_1 \cdot x_2 \cdots x_{n+1})$ is true,then at least one of $Q(x_1) \ldots Q(x_{n+1})$ must be true. Since the $x_1 \cdot x_{n+1}$ can be written as $(x_1 \cdot x_n) \cdot x_{n+1}$, so let a be $x_1 \cdot x_2 \cdots x_n$ and b be x_{n+1}, if $Q(a \cdot b)$ is true then one of $Q(a)$ or $Q(b)$ must be true.

- Case 1: if $Q(a)$ is true, then $Q(x_1 \cdots x_n)$ is true by the inductive hypothesis
- Cast 2: if $Q(b)$ is true, then $Q(x_{n+1})$ is true which proved the P$(n+1)$ is true directly.

So we proved $P(n)$ is true for all $n \geq 1$.

□